Designing mixed polyol cryoprotectants with solid–liquid equilibrium and a group-contribution activity model

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Theme: Thermodynamics in Biological Systems

Abstract

When a cell suspension freezes slowly, ice forms as pure water and the dissolved solutes concentrate in the shrinking liquid around the cells; this freeze concentration is a principal cause of freezing injury. Cryoprotectants are added to limit it, and are often combined because each becomes toxic at high concentration. This proposal predicts freeze concentration from solid–liquid equilibrium and uses the prediction to design a cryoprotectant mixture. Water activity in the unfrozen liquid is described by an excess Gibbs energy model in which the water–solute interaction is assigned per hydroxyl group, so that a single parameter fitted on glycerol predicts propylene glycol, sorbitol and their mixtures. The predictions are tested against freezing points measured with a low-cost cooling-curve setup, and the validated model is used to find the least-toxic glycerol + sorbitol recipe that keeps the residual salt concentration below a target at the storage temperature. Preliminary calculations show that a 10 wt% glycerol solution retains only 28% of its liquid at −10 °C, at 3.5 times its original concentration, and that a fixed-dose formulation of the design problem is independent of the thermodynamics — motivating the constrained formulation proposed here. [Add the fitted parameter and blind-test result once available.]

Keywords: cryopreservation; solid–liquid equilibrium; activity coefficients; group contribution; freeze concentration; mixture design



Graphical abstract. A single interaction parameter per –OH group is fitted on literature glycerol data, used to predict other polyols and their mixtures before they are measured, and then used to design a cryoprotectant recipe.

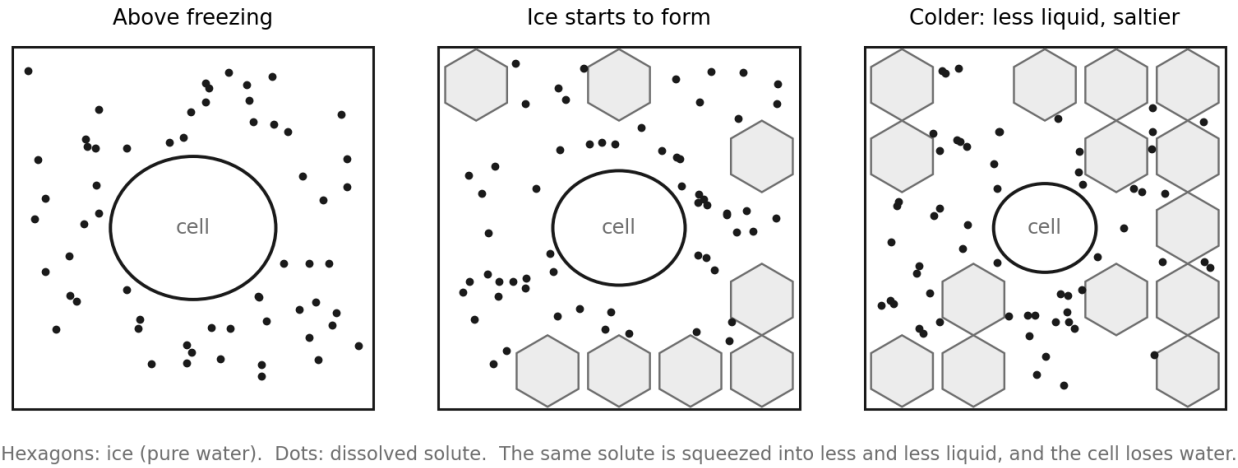


Figure 1: Freeze concentration. Ice excludes solute, so the same amount of solute occupies less liquid as temperature falls, and water is drawn out of the cell.

1 Introduction and background

The problem. Cells, tissues and biologics are stored frozen, but freezing damages them. In slow freezing, extracellular ice forms first; the remaining solution becomes progressively concentrated and water leaves the cells by osmosis. Prolonged exposure to this concentrated solution — *solution effects injury* — is one arm of the two-factor account of freezing damage [1]. Cryoprotectants such as glycerol reduce it; their classic role is to dilute the salt that would otherwise concentrate in the residual liquid [2]. Because each cryoprotectant is toxic at high concentration, mixtures are widely used to share the dose.

Alignment with the theme. The composition of the residual liquid at any temperature is fixed by phase equilibrium between ice and solution, not by kinetics or by the cell. Selecting a cryoprotectant and its concentration is therefore, in the first instance, a solid–liquid equilibrium problem — squarely within the thermodynamics of biological systems.

Existing approaches. Phase diagrams of water with common cryoprotectants have been measured since the 1960s [3], and cryobiology describes non-ideal solutions with the osmotic virial equation [4]. Separately, food science has applied group-contribution methods (UNIFAC) to water activity in aqueous polyols [5, 6], and extracts freezing behaviour from cooling curves [7]. A glycerol–water freezing experiment also exists as an undergraduate laboratory [8].

Relevant principles. At equilibrium the fugacity of water is equal in ice and in the solution. With the fusion free energy obtained from a heat–melt–cool construction [9],

$$\ln(x_w \gamma_w) = -\frac{\Delta_{\mathrm{fus}} H}{R}\left(\frac{1}{T} - \frac{1}{T_m}\right), \tag{1}$$

where the activity coefficient γ_w follows from an excess Gibbs energy model. Equation (1) has a consequence central to this work: once ice is present, the water activity of the residual liquid — and hence its osmotic pressure — is fixed by temperature alone. A cryoprotectant cannot change the osmotic stress at a given temperature; it changes how much liquid remains and what that liquid contains.

Gap and contribution. The building blocks above are established, but they sit in different fields. The contribution proposed here is to (i) test whether the food-science group-contribution approach transfers to cryoprotection conditions, (ii) predict mixed-cryoprotectant freezing behaviour blind, from single-component parameters, and (iii) use the validated model to *design* a mixture under a toxicity constraint, then verify the recommended recipe experimentally. Step (iii) was not found in this form in the literature searched [complete Google Scholar check before Checkpoint 2].

2 Proposed methodology

2.1 Activity model with per-group interactions

Only water–solute pair interactions are retained:

$$\frac{G^{\mathrm{ex}}}{RT} = \sum_i \alpha_i x_i x_w, \qquad \alpha_i = \frac{n_{\mathrm{OH},i}\, a_{\mathrm{OH}}}{RT}, \qquad \ln \gamma_w = (1 - x_w) \sum_i \alpha_i x_i. \tag{2}$$

For one solute this reduces to one-constant Margules. The single parameter a_{OH} is fitted by least squares on literature glycerol freezing points, with bootstrap uncertainty and a numerical Gibbs–Duhem check. Freezing points follow from solving (1) with (2); the residual liquid at temperature T follows from a solute mass balance, and the osmotic pressure from $\Pi = -RT \ln a_w / \bar{V}_w$.

2.2 Blind prediction

Using a_{OH} from glycerol alone, freezing curves are predicted for propylene glycol (2 –OH), sorbitol (6 –OH) and glycerol + sorbitol mixtures. Predictions are committed to a version-controlled repository before any measurement, providing a timestamped record that the test was blind. Systematic deviations are diagnostic: error growing with molecular size indicates a size effect (Wilson model); error confined to high concentration indicates approach to the glass-forming region.

2.3 Recipe design

Both doses w_{gly} and w_{sorb} are varied. The recipe minimises a toxicity cost $(w_{\mathrm{gly}}/c_{\mathrm{gly}})^2 + (w_{\mathrm{sorb}}/c_{\mathrm{sorb}})^2$ subject to the NaCl mass fraction of the residual liquid at the storage temperature not exceeding a target. Tolerable concentrations c_i and the salt target are taken from the literature [values and citations]. The medium is 0.9 wt% saline, with NaCl treated as ideal ions.

2.4 Experiment

Freezing points are measured by the cooling-curve method (Fig. 2). Samples are made by weight, cooled in an ice–salt bath, and logged every 2 s by DS18B20 probes read by an Arduino. The freezing point is taken as the maximum after recalescence, since a solution's plateau slopes downward as it freeze-concentrates; seeding limits supercooling. The rig is calibrated on pure water and on NaCl solution of tabulated freezing point. The run plan comprises calibration (4 runs), a glycerol check against literature (4), propylene glycol and sorbitol at 10, 20 and 30 wt% (12), three glycerol:sorbitol ratios (6), and two runs at the recommended recipe (2), all in duplicate.

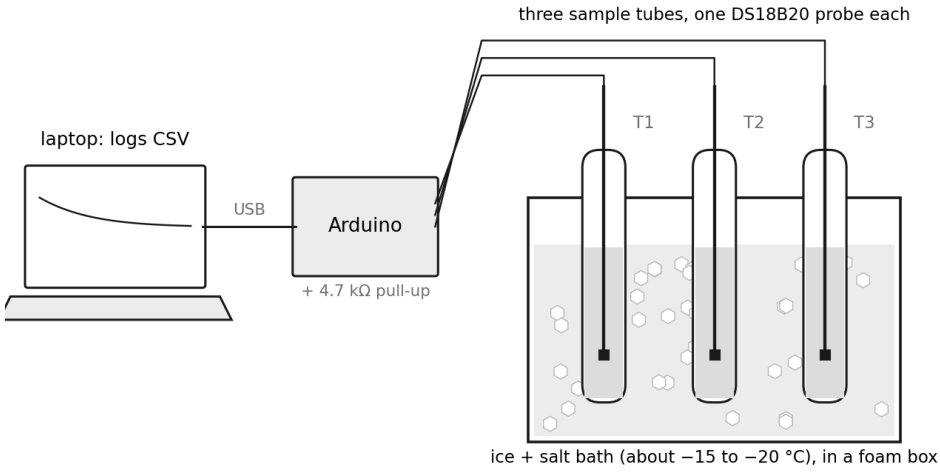


Figure 2: Experimental setup: three samples logged simultaneously in an ice–salt bath.

3 Preliminary results and discussion

3.1 Rough calculations

Table 1 gives results computed directly from equation (1) with $\Delta_{\text{fus}}H = 6.01\,\text{kJ mol}^{-1}$ and $\gamma_w = 1$. They show the scale of freeze concentration: a sample starting at 10 wt% glycerol is less than a third liquid at −10 °C, and that liquid is over three times as concentrated.

Table 1: Freeze concentration of a 10 wt% glycerol solution (ideal-solution estimate).

Temperature	−5 °C	−10 °C	−15 °C
Residual liquid concentration	20.5 wt%	35.1 wt%	45.9 wt%
Fraction of sample still liquid	48.7%	28.5%	21.8%
Osmotic pressure of residual liquid	6.1 MPa	12.2 MPa	18.3 MPa

3.2 A finding from formulating the design problem

If the total cryoprotectant dose is fixed and the aim is to minimise the larger of the two normalised exposures, the optimal ratio equals $c_{\text{gly}}/(c_{\text{gly}} + c_{\text{sorb}})$ for *any* activity model: both solutes are conserved in the residual liquid, so their ratio never changes. Thermodynamics drops out entirely. The constrained formulation of Section 2.3 avoids this, and on test inputs the ideal and non-ideal models recommend measurably different recipes.

3.3 Model fit and blind test

Figure 3 — to be generated from real data.
Freezing point against concentration for propylene glycol, glycerol and sorbitol: model curves (fitted on glycerol only) with measured points and error bars.
Table 2 — fitted a_{OH} with 95% confidence interval, RMSE, Gibbs–Duhem residual.

3.4 Recipe

Figure 4 — to be generated from real data.
Feasible recipes meeting the salt target, the minimum-toxicity recipe for the ideal and fitted models, and the measured freezing point of the recommended recipe.

3.5 Practicability

All materials are food or pharmaceutical grade and low-hazard; the setup costs little and can be run outside a dedicated laboratory. Measurement accuracy is limited by probe calibration (±0.1 °C after calibration against water) and by the sensitivity of the fitted parameter at low concentration, where a 0.2 °C error changes the Margules constant by about $10\,\text{kJ mol}^{-1}$; the fit therefore relies on the full concentration range. The equilibrium model does not apply below the glass transition, and the eutectic lies below the reach of the apparatus, so it is reported as a prediction only. [Discuss results.]

4 Conclusion and next steps

Potential impact. The proposal turns a phase diagram into a decision: which mixture of cryoprotectants to use, and at what doses, for a given storage temperature. Optimising freezing protocols empirically requires varying agent, concentration, cooling rate and cell type at once, with each survival assay taking days; a model calibrated on a few freezing points covers the composition axis continuously and reduces that search to confirmatory experiments. The same framework applies to lyophilised vaccine and biologic formulations, where the freezing curve and glass transition govern stability, and to freeze concentration in food processing.

Scope. The model predicts the environment a cell experiences, not its survival; the quantitative link between residual concentration and survival remains contested [1].

Next steps.

1. Couple the predicted residual concentration to a membrane water-transport model to predict cell volume during cooling, and hence an optimal cooling rate for a specific cell type.
2. Extend the activity model with a size term (Wilson) if the blind test shows a size effect.
3. Add solute–solute interactions if mixture predictions deviate systematically.
4. Compare with the osmotic virial equation used in cryobiology on the same data.

Code and data. All models, fitting, analysis and logging code, with raw data, are in a public repository [GitHub link].

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